

Project Initialization and Planning Phase

Date	5 July 2026
Team ID	
Project Title	SmartLender - Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	To develop an intelligent loan approval system that uses machine learning to detect fraudulent loan applications and evaluate applicant creditworthiness, enabling faster, more accurate, and secure lending decisions
Scope	The project automates loan application processing, document verification, fraud detection, and loan approval. It provides a secure, scalable, and user-friendly platform for applicants and lending institutions.
Problem Statement	
Description	Traditional loan approval systems rely heavily on manual verification, making them time-consuming and vulnerable to fraud. Detecting fake documents and fraudulent applicants is challenging, leading to financial losses and delayed loan processing.
Impact	Solving this problem will reduce fraudulent loan approvals, improve processing speed, enhance customer satisfaction, and increase the efficiency and security of financial institutions.
Proposed Solution	

Approach	Develop a web-based Smart Lender system that applies machine learning algorithms to analyze applicant data, verify documents, calculate fraud risk scores, and support loan approval decisions in real time.
Key Features	• Secure user registration and login. • Online loan application submission. • AI-based fraud detection using machine learning. • Automated document verification. • Creditworthiness assessment. • Real-time loan approval/rejection recommendations. • Admin dashboard for monitoring applications and fraud reports. • Email/SMS notifications for application status updates.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690